INCH-POUND

MIL-DTL-53084C
w/AMENDMENT 1
01 May 2026
SUPERSEDING
MIL-DTL-53084C
18 September 2023

DETAIL SPECIFICATION

PRIMER, CATHODIC ELECTRODEPOSITION, CHEMICAL AGENT RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers waterborne, cathodic epoxy electrodeposition primer formulated free of lead, hexavalent chromium and hazardous air pollutants (HAPs). The primer meets solvent emission maximums of 144 grams/liter (g/l) (1.2 pounds/gallon (lbs/gal)) volatile organic compound (VOC) content and is compatible with Chemical Agent Resistant Coating (CARC) topcoats. This primer is intended for use on properly cleaned and pretreated ferrous and nonferrous metal surfaces.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Director, U.S. Army, DEVCOM Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, ATTN: FCDD-RLA-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information may change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings).

FEDERAL STANDARDS

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-DTL-12468	-	Decontaminating Agent, STB.
MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant.

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY (EPA)

The Clean Air Act Amendments of 1990 List of Hazardous Air Pollutants

(Copies of this document are available online at https://www.epa.gov/ttn/atw/orig189.html.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM A109/ A109M	-	Standard Specification for Steel, Strip, Carbon (0.25 Maximum Percent), Cold-Rolled.
ASTM A1008/ A1008M	-	Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable.
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus.
ASTM D522	-	Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings.
ASTM D523	-	Standard Test Method for Specular Gloss.
ASTM D1153	-	Standard Specification for Methyl Isobutyl Ketone.
ASTM D1394	-	Standard Test Methods for Chemical Analysis of White Titanium Pigments.
ASTM D1475	-	Standard Test Method for Density of Liquid Coatings, Inks, and Related Products.
ASTM D1654	-	Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments.
ASTM D2196	-	Standard Test Methods for Rheological Properties of Non-Newtonian Materials by Rotational (Brookfield type) Viscometer.
ASTM D2369	-	Standard Test Method for Volatile Content of Coatings.
ASTM D3335	-	Standard Test Method for Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy.
ASTM D3359	-	Standard Test Methods for Measuring Adhesion by Tape Test.
ASTM D3363	-	Standard Test Method for Film Hardness by Pencil Test.
ASTM D3924	-	Standard Specification for Standard Environment for Conditioning and Testing Paint, Varnish, Lacquer and Related Materials.
ASTM D3960	-	Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings.
ASTM D4399	-	Standard Test Method for Measuring Electrical Conductivity of Electrocoat Baths.
ASTM G50	-	Standard Practice for Conducting Atmospheric Corrosion Tests Metals.

(Copies of these documents are available from https://www.astm.org).

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GENERAL MOTORS CORPORATION

GMW14872 - Cyclic Corrosion Laboratory Test.

(Copies of this document are available from https://global.ihs.com).

SAE INTERNATIONAL STANDARDS / Aerospace Material Specifications

SAE-STD-595 - Colors Used in Government Procurement

SAE-STD-595/26270 - Gray, Semigloss (Haze).
SAE-STD-595/26373 - Gray, Semigloss (Light).
SAE-STD-595/26493 - Gray, Semigloss (Pearl).
SAE-STD-595/37030 - Miscellaneous, Flat or Lusterless (Black).

(Copies of these documents are available online at https://www.sae.org).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The primer furnished under this specification shall be a product that is authorized by the qualifying activity for listing on the applicable Qualified Products Database (QPD) before contract award (see 4.2 and 6.3). Any change in the formulation of a qualified product shall necessitate its requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the products receiving qualification.

3.2 Materials. The materials used in the primer shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3 Color and 60º gloss. For qualification purposes, the primer shall be a gray visual color match to chip number 26493 of SAE AMS-STD-595, or at the most no darker than chip number 26373. Qualification is also acceptable for black, color chip number 37030. For actual production, the color may darken to chip number 26270. When black primer is applied, the CARC topcoats conforming to MIL-DTL-53039 or MIL-DTL-64159 shall be applied at a minimum dry film thickness of 2.0 mils (50 microns) to ensure the infrared reflectance (IR) signature requirements of the topcoats are met. The 60º gloss range shall be 30 - 85.

3.4 Composition. Cathodic electrocoat primer can be supplied in a single container as premixed resin and pigment paste, or in two separate containers, one for the resin and another for the pigment paste. The composition for the primer shall be as follows.

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3.4.1 Resin feed component. The resin feed component shall consist of an epoxy-based or an epoxy-urethane resin combined with the necessary amounts of flow control agents and volatile solvents. The use of acrylic resins as flow control agents is restricted to less than 2.0 percent by weight. When tested as specified in 4.8, the resin feed component shall meet the quantitative requirements of table I.

3.4.2 Pigment paste component. The pigment paste shall consist of an epoxy-based resin, volatile solvents, titanium dioxide (TiO_2) and/or carbon black pigments, and siliceous extenders. The use of small amounts of tinting pigments is permissible to achieve the color as specified in 3.3. Hexavalent chromate, zinc chromate, or lead pigments shall not be used alone or as a component part of any pigment. Compounds of antimony, arsenic, beryllium, cadmium, chromium, cobalt, cyanide, lead, manganese, mercury, nickel and selenium shall be absent. When tested as in section 4, the pigment paste component shall meet the quantitative requirements of table I.

3.4.3 Single container (one component) feed. The one component feed shall consist of the resin feed component specified in 3.4.1 and the pigment paste component specified in 3.4.2 blended together. When tested as in section 4, the single container feed shall meet the quantitative requirements of table II.

TABLE I. Component breakdown (two component systems).

Characteristic	Resin Feed	Gray Pigment Paste	Black Pigment Paste
Total solids, percent by weight	38.0 ± 5.0	53.0 ± 2.0	40 - 55
Weight, lbs/gal	8.8 ± 0.2	11.6 - 13.5	8.5 - 12.0
pH range	5.5 - 7.0	6.7 - 8.0	6.7 - 8.0
Viscosity, centipoise (cP)	250 cP (max)	5000 cP (max)	5000 cP (max)
TiO_2, percent by weight of pigment	-	50 (min)	N/A
Hexavalent chromium	-	Negative	Negative
Epoxy-based resin	Positive	-	-

TERMS: 1 centipoise (cP) = 0.01 poise (P).

TABLE II. Component breakdown (one component systems).

Characteristic	Cationic Feed (CF)
Total solids, percent by weight	34 - 46
Weight, lbs/gal	9.0 ± 0.5
Viscosity, centipoise (cP)	100 - 2500 cP
Hexavalent chromium	Negative
Epoxy resin	Positive

TERMS: 1 centipoise (cP) = 0.01 poise (P).

3.5 Mixed primer requirements. The mixed primer shall conform to the quantitative requirements of table III for a two component system when tested as in section 4. The mixed primer shall conform to the quantitative requirements of table IV for a one component system when tested as in section 4.

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TABLE III. Mixed primer quantitative requirements (two component systems).

Characteristic	Requirement (Gray)	Requirement (Black)
Total solids, percent by weight	10.0 - 20.0	10 - 23
Pigment to binder ratio	0.35 (max)	0.30 (max)
pH range	5.6 - 6.3	5.0 - 6.5
Conductivity, micromhos	1000 - 2000	850 - 2300
VOC content, g/l (lbs/gal)	144 (1.2) (max)	144 (1.2) (max)
Lead content, percent by weight of total solids	0.06 (max)	0.06 (max)

3.6 Applied primer qualitative requirements.

3.6.1 Condition in container. When tested as specified in 4.9.1, the resin feed component and pigment paste component shall be free from grit, seeds, abnormal thickening or livering in a freshly opened container. The pigment paste component shall show no more pigment settling or caking than can be easily and completely reincorporated to a smooth homogeneous state.

TABLE IV. Mixed primer quantitative requirements (one component systems).

Characteristic	Requirement
Total solids, percent by weight	10 - 23
pH range	4.8 - 6.5
Conductivity, micromhos	850 - 2300
VOC content, g/l (lbs/gal)	144 (1.2) (max)
Lead content, percent by weight of total solids	0.06 (max)

3.6.2 Mixing properties. When mixed as specified in 4.9.2, a smooth homogeneous mixture shall result.

3.6.3 Electrocoating application properties. When applied as specified in 4.9.3, the primer shall electrocoat satisfactorily and shall show a continuous film after baking which conforms to the color, gloss and performance properties as specified in section 4. The baked film shall show no mottling or color separation and shall present a finish free from craters or orange peel.

3.6.4 Cure time. When tested as specified in 4.9.4, the tested area shall show no film softening when compared to an untested portion of the panel.

3.6.5 Adhesion. When tested as specified in 4.9.5, the film of the primer shall show no removal from the surface of the cross-cut area, classification 5B in accordance with ASTM D3359. The film of the primer and the topcoat shall show less than 5 percent of the area affected as described as classification 4B in accordance with ASTM D3359.

3.6.6 Flexibility. A film of the primer tested as specified in 4.9.6 shall withstand bending without cracking or flaking.

3.6.7 Corrosion resistance.

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3.6.7.1 Salt spray resistance. When tested as specified in 4.9.7.1, the panels shall be examined immediately after removal from the salt spray cabinet. Rinse the panels and examine for rusting. The panels shall show minimal blistering or loss of adhesion of the paint from the scribe mark (for steel, a minimum rating of 6, for aluminum, a minimum rating of 8, ASTM D1654 Procedure A, Method 1) (see 4.9.7.1). Blisters on the non-scribed areas shall cover no more than 5% of the exposed area, none of which shall be larger than 1mm in diameter (rating of 7; see table V. Ratings for nonscribed coatings). The panels tested for cross hatch adhesion properties shall satisfy classification 3B of ASTM D3359. Upon removal of the primer, the substrate shall show no more than a trace of rusting, pitting, or corrosion.

TABLE V. Ratings for nonscribed coatings.

Area Failed, %	Rating Number
No Failure	10
0 to 1	9
2 to 3	8
4 to 6	7
7 to 10	6
11 to 20	5
21 to 30	4
31 to 40	3
41 to 55	2
56 to 75	1
Over 75	0

3.6.7.2 Cyclic corrosion resistance. A pretreatment with a film of the primer tested as specified in 4.9.7.2 and evaluated using ASTM D1654 Procedure A, Method 1 shall have a minimum rating of seven (7). Blisters shall cover no more than 5% of the exposed area, none of which shall be larger than 1mm in diameter (rating of 7; see table V, Ratings for nonscribed coatings).

3.6.7.3 Atmospheric corrosion resistance. A pretreatment with a film of the primer and topcoat tested as specified in 4.9.7.3 shall show minimal blistering or loss of adhesion of the paint from the scribe mark (for steel, a minimum rating of 6, for aluminum, a minimum rating of 8, ASTM D1654 Procedure A, Method 1). Blisters on the non-scribed areas shall cover no more than 5% of the exposed area, none of which shall be larger than 1mm in diameter (rating of 7; see table V, Ratings for nonscribed coatings).

3.6.8 Super tropical bleach (STB) resistance. When tested as specified in 4.9.8, a film of the coating shall show no blistering, wrinkling, or film softening when examined immediately after washing with water. Film softening shall not exceed a 2 pencil hardness difference (see ASTM D3363) from an unexposed film with identical cure history prior to STB exposure. After drying there shall be a maximum color change of 2.5 NBS units when comparing a portion of the untested panel to that of the tested area. The STB composition shall be in accordance with MIL-DTL-12468.

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3.6.9 Throw power. When tested as specified in 4.9.9, the inside face of the test panel shall plate to a dry film thickness of 0.1 ± 0.01 mils (2.5 ± 0.25 microns) minimum at least 10 inches (254 mm) up the panel. A throw power below 10 inches (254 mm) is unacceptable. Blisters above the 10-inch mark shall cover no more than 5% of the exposed area, none of which shall be larger than 1mm in diameter (rating of 7: see table V, Ratings for nonscribed coatings).

3.6.10 Toxic ingredients. Other than parachlorobenzotrifluoride (PCBTF) and ethylene glycol n-butyl ether (butyl cellosolve), the primer shall contain no benzene, chlorinated solvents or ethylene based glycol ethers and their acetates (see 4.9.10). The primer shall have no adverse effects on the health of personnel when used for its intended purpose.

3.6.11 HAP ingredients. The primer shall be formulated free of HAPs (see 4.9.11). The current list of HAPs can be found on The Clean Air Act Amendments of 1990 List of Hazardous Air Pollutants (see 2.2.2).

3.6.12 User instruction marking and precaution sheet. All containers shall include the VOC content in g/l of coating and a printed precaution sheet with the following information:

PRECAUTION: The Surgeon General requires airline respirators to be used unless air sampling shows exposure to be below standards. Then, either chemical cartridge respirators or airline respirators are required. Avoid contact with skin and eyes. Use with adequate ventilation. For other safety recommendations, refer to the Safety Data Sheet (SDS). Keep containers closed.

3.6.13 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief.

See:

(https://www.osha.gov/dsg/hazcom/) and as specified in Appendix D of 29 CFR 1910.1200.

and:

(https://www.osha.gov/laws-egs/regulations/standardnumber/1910/1910.1200AppD).

A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the qualifying activity (see 6.3.2). The SDS shall be included with each shipment of the material covered by this specification and submitted to pertinent Government agencies as stated in FED-STD-313.

3.6.14 Toxicity clearance. All new chemicals and materials being added to the Army supply system shall have a toxicity clearance. A toxicity clearance involves a toxicological evaluation of materials prior to introduction into the Army supply system. The Army program manager shall be responsible for identifying technically feasible materials and requesting a toxicity clearance for use of that material within their program (see 6.3.4).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification shall be conducted by the qualifying activity (see 6.3). The samples shall be legibly identified (see 6.3.3). Qualification inspection shall consist of tests for all requirements in section 3 in accordance with table VI, and examination for user instruction marking (see 3.6.12). The results of each test shall be compared with the applicable requirement in section 3. Failure to conform to any requirement shall be counted as a defect, and paint represented by the sample test panels and disks shall not be approved for inclusion on the QPD under this specification.

4.3 Conformance inspection. The manufacturer shall perform conformance inspection testing on each production lot (see 4.3.1) and have these test results on file when requested by the contracting officer or the procuring activity. Conformance inspection shall consist of testing for the requirements as specified in table III or table IV depending upon the component system. There shall be no failures (see 6.4).

4.3.1 Lot and batch formation. Unless otherwise specified in the contract or purchase description (see 6.2), a lot shall consist of all coatings of the same type, composition and color, from a single uniform batch, produced and offered for delivery at one time (see 6.3.1). Unless otherwise specified in the contract or purchase description (see 6.2), a batch shall consist of all coating material (in U.S. gallons) manufactured during one continuous operation and forming part of one contract or order for delivery (see 6.3.1). The addition of any substance to a batch shall constitute a new lot.

4.4 On-Site production tests. On a monthly basis, under production conditions, the contractor shall electrocoat and cure a standard pretreated panel. The tank operating conditions at the time of application shall be noted on the panel (date, voltage, tank temperature, line speed, dry film thickness, bake temperature, bake time). The panel, along with a gallon tank sample, shall be sent to the coating manufacturer for evaluation. The panel, test results and analyses of the tank sample shall be retained by the coating supplier for one year. This data shall be presented to responsible parties in the Department of Defense upon request.

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with ASTM D3924. Also, unless otherwise stated in the test method or paragraph, room temperature shall be 73 ± 4 ºF (23 ± 2 ºC) and a 40 - 70 percent relative humidity. In addition to ASTM D3924, the following determinations shall be done at the primer application line:

a. Daily tests for percent solids, pH and conductivity of the primer bath.

b. Daily tests for conductivity of the deionized rinse water, post rinse before the primer bath and the anolyte.

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c. One gallon samples of the primer bath shall be evaluated on a monthly basis by the coating supplier (see 4.4).

This data shall be presented to responsible parties in the Department of Defense upon request.

4.5.1 SDS. The SDS shall address all components of the primer and shall comply with the requirements of FED-STD-313. Nonconformance to 3.6.13 shall constitute failure of this requirement.

4.6 Test methods.

4.6.1 Test conditions. Except as otherwise specified herein, the routine and referee testing shall be conducted in accordance with ASTM D3924. A dry film thickness of 1.0 ± 0.2 mils (25 ± 5 microns) shall be used whenever film thickness is requested in any test, unless otherwise required by the test. Failure of any test result to fall within the specified ranges in section 3 shall constitute failure of the applicable test. For all tests requiring the use of the mixed primer, the resin feed component and pigment paste feed component shall be mixed in the proportions specified in 4.6.3.

4.6.2 Test panels. Steel test panels shall be cold-rolled SAE 1010 steel conforming to ASTM A109/A109M or ASTM A1008/A1008M, treated with a zinc phosphate coating in accordance with TT-C-490, type I. The panels shall have a Rockwell B hardness (HRB) of 55 to 75 and a surface roughness of 30 to 45 microinches (arithmetic average) as rolled. Aluminum test panels shall be aluminum alloy 2024-T3 treated with MIL-DTL-5541, type II.

4.6.3 Mixing and application. Unless otherwise specified (see 6.2), instructions for mixing and thinning the mixed primer shall be specified in the contract or solicitation. In preparation of the primer bath, the deionized water used for thinning and rinsing shall have a conductivity of less than 10 micromhos/centimeter and be free of bacteria. Conductivity shall be measured in accordance with ASTM D4399. After mixing the resin, pigment and deionized water, allow the bath to stir for a 24 hour period before testing. Throughout the testing period, constantly agitate the bath. The bath operations shall be held between 88 - 92 ºF (31 - 33 ºC), or at other temperatures according to the supplier's recommendation. Since the electrodeposition primer is cathodic, the panel being coated is the negative electrode while the beaker or side electrodes are positively charged. Panels to be electrocoated are immersed in the bath liquid and the rectifier turned on. Voltage is increased over approximately 20 seconds to a reading necessary to provide 1.0 ± 0.1 mils (25 ± 2.5 microns) of dry film thickness. Several trial panels are necessary to define this voltage for each substrate. Generally, 150 to 300 volts are required. Normal deposition time is 90 seconds. Once coated, panels are removed from the bath and rinsed with deionized water. Curing of the deposited primer shall be in a convection type oven capable of reaching 350 ± 2 °F (177 ± 1 ºC) metal temperature for 20 minutes, or alternative curing schedule as recommended by the coating supplier. Allow the cured primer to cool to room temperature 65 - 80 °F (18 - 27 ºC) before testing.

4.6.4 Test procedures. Tests (see table VI) shall be conducted in accordance with the appropriate ASTM method, or as specified herein. The right is reserved to make any additional tests deemed necessary to determine that the coating meets the requirements of this specification.

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4.7 Color and 60º gloss. Prepare a panel of each substrate as specified in 4.6.2 and 4.6.3. Compare the color as specified in 3.3. Determine the 60º gloss in accordance with ASTM D523 and check for compliance to 3.3. Nonconformance to 3.3 shall constitute failure of this test.

TABLE VI. Index.

Item	ASTM Method	Test Paragraph	Requirement Paragraph or Table
Color and 60º gloss	D523	4.7	3.3
Resin feed component	-	4.8	3.4.1
Pigment paste component	-	4.8	3.4.2
Total solids	D2369	4.8.1	Table I, II, III and IV
Weight/gallon	D1475	4.8.2	Table I and II
pH range	-	4.8.3	Table I, III and IV
Viscosity	D2196 Method A	4.8.4	Table I and II
Epoxy resin test	-	4.8.5	Table I, II and 3.4.1
TiO_2	D1394	4.8.6	Table I
Hexavalent chromium	-	4.8.7	Table I, II and 3.4.2
Pigment to binder ratio	-	4.8.8	Table III
Conductivity	-	4.8.9	Table III and IV
VOC content	D3960	4.8.10	Table III and IV
Lead content	D3335	4.8.11	Table III and IV
Condition in container	-	4.9.1	3.6.1
Mixing properties	-	4.9.2	3.6.2
Electrocoating application properties	-	4.9.3	3.6.3
Cure time	D1153	4.9.4	3.6.4
Adhesion	D3359 Method B	4.9.5	3.6.5
Flexibility	D522 Method B	4.9.6	3.6.6
Salt spray resistance	B117	4.9.7.1	3.6.7.1
Cyclic corrosion resistance	GMW 14872	4.9.7.2	3.6.7.2
Atmospheric corrosion resistance	ASTM G50	4.9.7.3	3.6.7.3
STB resistance	-	4.9.8	3.6.8
Throw power	-	4.9.9	3.6.9
Toxic ingredients	-	4.9.10	3.6.10
HAP ingredients	-	4.9.11	3.6.11

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4.8 Composition test methods. The following tests shall be made on the resin feed component, pigment paste component, single container (one component) feed and mixed primer as indicated in tables I, II, III, and IV.

4.8.1 Total solids. Determine the total solids (nonvolatile matter) in accordance with ASTM D2369.

4.8.2 Weight/gallon. Determine the weight/gallon in accordance with ASTM D1475.

4.8.3 pH determination. Standardize pH meter using fresh buffer solutions with pH of 4, 6, and 7 at 77 °F (25 °C). Rinse the pH probe with distilled or deionized water between each reading. Immerse the probe into the sample and take the reading after one minute with mild agitation.

4.8.4 Viscosity. Determine the viscosity in accordance with ASTM D2196 test method A.

4.8.5 Epoxy resin test. Determine the presence of an epoxy-based resin by placing approximately 5 ml of the well mixed resin component into a test tube. Extract the sample with 10 ml of toluene. Decant most of the top organic layer into a 25 ml beaker and evaporate to dryness on a steam bath. After cooling, add dropwise 2 ml of concentrated H_2SO_4. Stir, cool and add 2-3 drops of 40 percent formaldehyde solution. Allow sample to stand a few minutes. Dilute with 15 ml of distilled water added all at one time. A green to blue-green color will form immediately indicating the presence of a bisphenol type of epoxy resin. Confirmation of the resin type shall be determined by placing two drops of the organic layer extract onto a KBr plate. Remove the solvent by drying the plate in a 140 °F (60 °C) oven for 30 minutes. Identify the remaining film by infrared spectroscopy. Nonconformance to tables I and II shall constitute failure of the test.

4.8.6 TiO_2 content. Determine the TiO_2 content on the extracted pigment in accordance with ASTM D1394.

4.8.7 Hexavalent chromium. Determine the presence or absence of hexavalent chrome by using pigment extracted as in 4.8.6. Add 5 ml of 25 percent aqueous KOH to 0.5 g of the extracted pigment in a 15 ml centrifuge tube. Agitate by shaking the tube for a few minutes then centrifuge. The supernatant liquid shall be colorless. A yellow color indicates the presence of hexavalent chrome and shall constitute failure of this test.

4.8.8 Pigment to binder ratio. Determine the pigment to binder ratio as an average of two sample test results. Precondition porcelain crucibles in an oven at 230 ± 9 °F (110 ± 5 °C) for 30 minutes and store in a desiccator prior to use. Weigh the crucibles and record the weight, C. Transfer 1.0 to 1.5 g of the mixed primer bath into the crucible and bake for one hour at 230 ± 9 °F (110 ± 5 °C). After cooling the sample to room temperature in a desiccator, reweigh the crucible, D. Transfer the crucible to a muffle furnace preset to 212 °F (100 °C) or less and adjust the setting in order to reach a final temperature of 1220 °F (660 °C). The sample shall remain at 1220 °F (660 °C) for 90 minutes. Remove sample from muffle furnace and allow it to return to room temperature in a desiccator. Reweigh crucible and record the weight, E. Calculate the pigment to binder ratio as follows:

$$P/B = (E-C) / (D-E)$$

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4.8.9 Conductivity. Determine the conductivity of the mixed primer bath with a conductivity meter with a range of 0.1 to 10,000 micromhos/centimeter and by using a cell with a constant of 1.0. Before determining the conductivity of the sample, standardize the conductivity cell by using a 0.010 normal KCl solution. With a 0.010 normal KCl solution, having a conductivity reading of 1413 micromhos/centimeter at 77 °F (25 °C), the conductivity cell K factor is obtained as follows:

K factor = (1413) / (conductivity reading of KCl solution)

Disconnect the conductivity cup and discard the standard solution. Rinse the cup several times with deionized water and wipe dry. The test sample shall be homogeneous and at a temperature of 77 ± 1 °F (25 ± 1 °C). Fill the conductivity cup with the sample, reconnect the cup to the conductivity meter and record the apparent conductivity reading, A. Calculate the bath conductivity of the sample as follows:

Conductivity of mixed primer bath = (K factor) x A

4.8.10 VOC content. Determine the VOC content of the mixed primer bath in accordance with ASTM D3960.

4.8.11 Lead content. Determine the lead content of the mixed primer bath by atomic absorption spectroscopy in accordance with ASTM D3335.

4.9 Applied primer qualitative requirements.

4.9.1 Condition in container. Upon opening a full previously unopened container, the condition of the contents shall be examined for compliance with 3.6.1. Reseal, then agitate the container for three (3) minutes on a paint shaker and examine for compliance with 3.6.1. On re-examination of the contents, the disclosure of any gel bodies or undispersed pigment indicates unsatisfactory settling properties.

4.9.2 Mixing properties. Follow the manufacturer's instructions in preparing the primer bath. Charge the coating cell with deionized water as specified in 4.6.3 and under agitation slowly add the resin feed component. Gradually add the pigment feed under sufficient agitation to produce a homogeneous system. Check for compliance with 3.6.2. Nonconformance to 3.6.2 shall constitute failure of this test.

4.9.3 Electrocoating application properties. Deposit a film of the primer and cure as specified in 4.6.3. Check for compliance with 3.6.3. Nonconformance to 3.6.3 shall constitute failure of this test.

4.9.4 Cure time. Prepare a film of primer as specified in 4.6.3 ensuring that the primer is cured for 20 minutes at a metal temperature of 350 °F (177 °C) or according to the coating supplier’s recommendation. Allow the panel to cool to a room temperature of 65 - 80 °F (18 - 27 °C). Saturate a soft cloth or laboratory tissue with methyl isobutyl ketone conforming to ASTM D1153. Applying medium pressure with the index finger, rub the same area of the primer for a minimum of 30 strokes (movement in one direction). Examine the film of primer for loss of film and check for compliance with 3.6.4. Nonconformance to 3.6.4 shall constitute failure of this test.

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4.9.5 Adhesion. Prepare three panels of each substrate as specified in 4.6.3. Perform the cross hatch adhesion test on one panel of each substrate in accordance with ASTM D3359, method B. Check for compliance with 3.6.5. The remaining steel and aluminum panels shall be topcoated with MIL-DTL-53039 or MIL-DTL-64159 to a minimum dry film thickness of 1.8 mils (45 microns) and allowed to air dry for 168 hours. Perform the cross hatch adhesion test on each topcoated panel in accordance with ASTM D3359, method B. Check for compliance with 3.6.5. Nonconformance with 3.6.5 for either test shall constitute failure of this test.

4.9.6 Flexibility. Determine flexibility in accordance with ASTM D522, method B using a 1/4 inch mandrel. Prepare a film of primer on an aluminum panel as specified in 4.6.2 and 4.6.3. Bend the coated panels according to ASTM D522, method B. Examine the coating for cracks over the area of the bend for compliance with 3.6.6. Nonconformance to 3.6.6 shall constitute failure of this test.

4.9.7 Corrosion resistance.

4.9.7.1 Salt spray resistance. Prepare panels of each substrate as specified in 4.6.2 and 4.6.3. The painted specimen/coupons are to be "X" scribed using a tungsten carbide scribing tool as described in ASTM D1654 (paragraph 5.1.2 - pencil type). The "X" scribe is made by scribing two intersecting lines from one corner to its opposite corner across the face of the specimen/coupon. The scribe should initiate/terminate no closer than 1/2" to the edge of the specimen/coupon. The edges of all panels shall be sealed. The panels shall then be placed in a 5 percent salt spray cabinet for 1008 hours as described in ASTM B117. Upon removal, wash the panels gently in warm running or deionized (DI) water until free from any visible salt deposits and examine immediately for compliance with 3.6.7.1. One panel of each substrate shall have the cross hatch adhesion test performed in accordance with ASTM D3359, method B. Check for compliance with 3.6.7.1. One panel of each substrate shall be stripped down to the substrate and checked for compliance with 3.6.7.1. Nonconformance to 3.6.7.1 shall constitute failure of this test.

4.9.7.2 Cyclic corrosion resistance. Prepare panels of each substrate as specified in 4.6.2 and 4.6.3. The painted specimen/coupons are to be "X" scribed using a tungsten carbide scribing tool as described in ASTM D1654 (paragraph 5.1.2 - pencil type). Expose the specimens/coupons to accelerated cyclic corrosion for 30 cycles in accordance with GMW 14872, Ext, Exposure C. Upon removal, wash the specimens/coupons gently in warm running water or DI water until free from any visible salt deposits and examine immediately for compliance with 3.6.7.2. The specimens/coupons shall undergo coating removal along the scribe as per ASTM D1654 and inspected for rust, pitting or corrosion not evident with paint present. Nonconformance shall constitute failure of this test.

4.9.7.3 Atmospheric corrosion resistance. Prepare panels of each substrate as specified in 4.6.2 and 4.6.3. The painted specimen/coupons are to be "X" scribed through the primer and topcoat to the substrate. Expose these for two (2) years at a suitable test site in accordance with ASTM G50, with equivalent or greater mass loss than 1.5mpy such as the Army Research Laboratory (ARL) site at Cape Canaveral. Upon completion of the 2 year exposure, specimen/coupons shall be field evaluated, then returned to ARL for a final evaluation. Upon return, wash the specimens/coupons gently in warm running water or DI until free from any visible salt deposits and examine immediately for compliance with 3.6.7.3. The specimens/coupons shall undergo

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coating removal along the scribe as per ASTM D1654 and inspection for rust, pitting or corrosion not evident with paint present. Nonconformance shall constitute failure of this test.

4.9.8 Super tropical bleach (STB) resistance. Prepare a film of primer on steel as specified in 4.6.2 and 4.6.3. Scribe a 1 inch diameter wax ring using a china marker on the painted surface of the panel. Place approximately 1 ml of STB agent on the panel surface. Do not cover. Allow to stand 30 minutes then thoroughly wash with water. An STB slurry mix of 40 parts STB and 60 parts water by weight shall be used. Examine for compliance with 3.6.8. Nonconformance to 3.6.8 shall constitute failure of this test.

4.9.9 Throw power. In determining the throw power of the primer bath use 4 x 18 inch steel panels as specified in 4.6.2 and apply as specified in 4.6.3. Assemble a throw power box by securing the two metal panels face to face in a way that will result in a 3/8-inch gap between the panels. The panels shall be assembled along the sides using a nonconductive material with the top and bottom edges open. Submerge the throw power box into the primer bath and deposit a film of primer in accordance to 4.9.3. Application of the primer shall follow the parameters established in order to obtain 1.0 ± 0.1 mils (25 ± 2.5 microns) dry film deposited on the outside face of the panel to be coated. Cure the panels as specified in 4.9.4. Measure the film thickness at one inch intervals on the side of the panel which was located on the interior surface of the throw box. Seal the uncoated section and all edges of the test panel and expose the inside face to a 5 percent salt spray solution for 96 hours in accordance with ASTM B117. The panels shall not be scribed. Upon comp1etion of the exposure, remove the panels and wash in warm running water until they are free from any visible salt deposits. Examine immediately for compliance with 3.6.9. Nonconformance to 3.6.9 shall constitute failure of this test.

4.9.10 Toxic ingredients. Other than PCBTF and ethylene glycol n-butyl ether (butyl cellosolve), the manufacturer shall certify that the primer contains no benzene, chlorinated solvents or ethylene based glycol ethers and their acetates. Check for compliance with 3.6.10. Nonconformance to 3.6.10 shall constitute failure of this test.

4.9.11 HAP ingredients. The manufacturer shall certify that the primer contains no HAPs. Check for compliance with 3.6.11. Nonconformance to 3.6.11 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers a cathodic electrodeposition epoxy primer intended for use on properly cleaned and pretreated steel and aluminum. It is formulated free of lead, hexavalent chromium and HAPs. The primer has a maximum VOC level of 144 g/l (1.2 lbs/gal).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Required markings (see 3.6.13).
c. Whether a MSDS is required with each shipment (see 3.6.14 and 6.3.2).
d. If a toxicity clearance is required (see 3.6.15).
e. If qualification samples are required and where to send them (see 4.2 and 6.3).
f. If conformance samples are required and where to send them (see 4.3).
g. Lot and batch formation (see 4.3.1).
h. If instructions are required for mixing and thinning the mixed primer (see 4.6.3).
i. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the QPD, whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Samples for QPD testing (see 4.2) should be submitted to DEVCOM Army Research Laboratory, ATTN: FCDD-RLA-MC, Coatings and Corrosion Team, 6300 Rodman Road, Building 4600, Aberdeen Proving Ground, MD 21005-5066. Listings of qualified products can be obtained online through ASSIST, database at https://assist.dla.mil/.

6.3.1 Conformity to qualified sample. All lots of coatings supplied under this specification must be manufactured using the same formulation, raw materials and supplier(s) of raw materials, methods of manufacture, equipment, and geographic location as the qualification sample, unless changes have been approved by the qualifying activity.

6.3.2 SDS. Contracting officers will identify those activities requiring copies of a completed SDS prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.3.3 Sample identification for qualification inspection. Samples for QPD testing are to be identified in a cover letter with the following information:

a. Manufacturer's name and product number.

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b. Submitted by (name and date).
c. Specify the number of samples.
d. Specify the reason for submitting the samples.
e. Specification MIL-DTL-53084C Amendment 1; Color_____, "Primer, Cathodic Electrodeposition, Chemical Agent Resistant.
f. Provide a copy of the SDS.
g. Provide a copy of the notarized statement of composition.
h. Provide a copy of the technical data sheet.
i. Provide a copy of the test report.

6.3.4 Toxicity clearance request. Department of the Army Regulation (AR) 40-5, Preventive Medicine, (AR) 70-1, Acquisition Policy, and Department of the Army Pamphlet 70-3, Acquisition Procedures, require a toxicity clearance. Army toxicity questions and/or a toxicity clearance request should be addressed to:

US Army Public Health Command (USAPHC), ATTN: MCHB-IP-TTE, 5158 Blackhawk Road, APG-EA, MD 21010-5403 or emailed to: usaphctepinfo@amedd.army.mil

6.3.5 Retention of qualification. To retain qualification of products approved for listing in the QPD, the manufacturer will be requested to verify by certification to the qualifying activity that its product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in two-year intervals from the date of original qualification and will be initiated by the qualifying activity.

6.4 Conformance rejection and retest. Failure in any conformance inspection will result in the rejection of the batch from which it was obtained. Rejected material cannot be resubmitted for acceptance without written approval from the qualification activity (see 4.3). The application for resubmission will contain all details concerning previous rejections and measures taken to correct these deficiencies.

6.5 Detail specification. MIL-DTL-53072, Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection, is available for application procedures and quality control inspection of this coating.

6.6 Subject term (key word) listing.

Aluminum	Composition	Resin
CARC	Paste	Steel
Component	Pigment	Total solids

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS
Air Force - 11

Preparing activity:
Army - MR

(Project 8010-2025-009)

Review activities:
Army - MD1, MI
Navy - CG, SH
Air Force - 184

Civil agency:
GSA/FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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